



Machine learning prediction of childhood nephrotic syndrome outcomes

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Abstract

Background Children with steroid-resistant, frequently relapsing, and steroid-dependent nephrotic syndrome experience high disease and treatment-related morbidity. There are few prediction models for childhood nephrotic syndrome outcomes. Our aim was to develop and internally validate outcome prediction models, using machine learning methods.

Methods We analyzed data from Insight into Nephrotic Syndrome: Investigating Genes, Health, and Therapeutics, a prospective observational childhood nephrotic syndrome cohort. We included children (1–18 years) diagnosed from 1996 to 2023 from the Greater Toronto Area, Canada, with baseline data from within 90 days of diagnosis. Outcomes were frequent relapses or steroid dependence by 1 year, relapse occurrence by 1 year, steroid-sparing medication initiation by 2 years, and initial steroid resistance. We developed and compared the performance of nine different machine learning algorithms for each outcome. Predictors included sociodemographic, clinical, and laboratory features.

Results We included 515 children diagnosed with nephrotic syndrome. Of these, 484 (94%) were steroid-sensitive and 31 (6%) were steroid-resistant. Nine machine learning models were developed and optimized by hyperparameter tuning for each outcome. The random forest model had the best performance for predicting frequent relapses or steroid dependence, although its predictive ability was low (AUC 0.60). Machine learning models also had weak predictive ability for relapse occurrence (AUC 0.62; logistic regression with recursive feature elimination), steroid-sparing medication initiation (AUC 0.61; XGBoost), and steroid resistance (AUC 0.64; XGBoost).

Conclusions Routinely collected sociodemographic, clinical, and laboratory features at nephrotic syndrome diagnosis are weakly predictive of subsequent relapses and treatment response, using machine learning methods. Discovery of novel biomarkers may improve future prediction and proactive treatment.

Keywords Nephrotic syndrome · Children · Pediatric · Machine learning · Prediction model

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Introduction

Nephrotic syndrome is a common kidney disease frequently managed by pediatricians worldwide. The estimated incidence of childhood nephrotic syndrome is 2–7/100,000 children per year and is associated with substantial disease- and treatment-related morbidity [1–6]. After nephrotic syndrome diagnosis, up to 90% of children respond to glucocorticoid immunosuppression and are classified as steroid-sensitive [5–7]. However, 80% of children with steroid-sensitive nephrotic syndrome experience at least one disease relapse, and approximately half develop frequent relapses or steroid dependence [6, 7]. The chronic, relapsing disease course and high cumulative steroid exposure contribute to substantial disease- and treatment-related morbidity, diminished quality of life, and psychological distress among patients and their caregivers [8–10].

Previous studies suggest that age of onset, sex, and disease activity in the first 6 months may predict the frequency of subsequent nephrotic syndrome relapses [11–16]. Yet, there are currently a lack of validated clinical models or biomarkers able to predict steroid response, subsequent relapse patterns, or the need for steroid-sparing immunosuppressive medications from the time of disease onset. Previous clinical prediction models for frequent disease relapses have *C*-statistics ranging from 0.5 to 0.57, indicating poor discriminative performance [4]. An improved ability to predict steroid resistance and frequent disease relapses would facilitate earlier introduction of maintenance immunosuppressive therapies to induce remission or prevent relapses. This could reduce cumulative steroid burden and associated toxicity, identify high-risk patients warranting additional surveillance, and facilitate population enrichment for future randomized controlled trials. Few prediction models using machine learning methods have been developed in childhood nephrotic syndrome [17–20]. Prior models have included predictors not routinely measured in clinical practice (e.g., autoantibodies), measured after the initial nephrotic syndrome episode, and have developed models using methods that would result in information leakage and exaggerated performance. To address this knowledge gap and the limitations of prior studies, our aim was to determine whether patient sociodemographic, clinical, and laboratory features that are routinely collected at the time of childhood nephrotic syndrome diagnosis are predictive of subsequent nephrotic syndrome classification, relapse occurrence, and steroid-sparing immunosuppression treatment, using machine learning techniques.

Methods

Study design and population

Insight into Nephrotic Syndrome: Investigating Genes, Health, and Therapeutics (INSIGHT; ClinicalTrials.gov

identifier NCT1605266) [21] is a multi-center, prospective observational cohort study enrolling children (1–18 years) diagnosed with nephrotic syndrome after 1996 from the Greater Toronto and Hamilton Area, Canada. Nephrotic syndrome is defined using standardized Kidney Disease: Improving Global Outcomes (KDIGO) criteria [5]. The study received research ethics approvals at each clinical center and is reported according to the STrengthening the Reporting of OBservational studies in Epidemiology (STROBE) guidelines (Supplemental Appendix 1) [22]. For this analysis, we included all INSIGHT participants (1–18 years) diagnosed with nephrotic syndrome from 1996 to 2023 with an initial study visit within 90 days of diagnosis. This population overlaps with a previous study that developed clinical prediction models using non-machine learning methods among 631 children diagnosed with nephrotic syndrome between 1993 and 2016 [4]. Children with steroid-sensitive and steroid-resistant nephrotic syndrome were included. Children were excluded if they had monogenic nephrotic syndrome, secondary nephrotic syndrome (e.g., lupus nephritis), or biopsy-proven membranous nephropathy or membranoproliferative glomerulonephritis. Sociodemographic, clinical, laboratory, and questionnaire data are collected for INSIGHT participants from diagnosis. Study participants undergo annual follow-up until clinic discharge, typically 5 years after their last relapse or upon transition to adult care at age 18.

Predictors

We purposively selected a broad range of sociodemographic, clinical, and laboratory features, based on (1) their reported or theoretical associations with childhood nephrotic syndrome outcomes and (2) their availability in routine clinical practice around the time of nephrotic syndrome diagnosis (Supplemental Table 1). All predictor variables were collected from the time of diagnosis or enrolment study visit (within 90 days of nephrotic syndrome diagnosis). The period of 90 days was selected to maximize sample size and develop pragmatic prediction models based on data available during the initial nephrotic syndrome episode that could be applied at initial nephrology consultation visits. These included demographic factors (age, sex, rural residence, ethnicity, and immigration status), family socioeconomic factors (parental education, parental employment, parental income, and number of children in the household), birth history (birthweight and gestational age), family history (of nephrotic syndrome or other kidney diseases), atopic history (pre-existing asthma, eczema, seasonal allergies, and atopic disease severity by ISAAC score), baseline anthropometric and clinical features (body mass index, systolic and diastolic

blood pressure percentiles-for-age, month of diagnosis, and presenting symptoms including facial edema, peripheral edema, abdominal distension, weight gain, and low urine output), recent illness history (viral illness or infection treated with antibiotics within 6 months prior to diagnosis), initial management (hospitalization and intravenous albumin administration at diagnosis), and laboratory measures at diagnosis (proteinuria [urinalysis protein concentration, urine protein to creatinine ratio and albumin to creatinine ratio], microscopic hematuria, and serum laboratory measures [hemoglobin, albumin, cholesterol, vitamin D, sodium, estimated glomerular filtration rate, and urea]). We restricted to the closest laboratory measurements to the nephrotic syndrome diagnosis date within 2 weeks (before or after) to reflect measurements taken while the child was in the nephrotic syndrome state.

Outcomes

Among children with steroid-sensitive nephrotic syndrome, the primary outcome of interest was the binary classification of nephrotic syndrome relapse pattern by 1 year of diagnosis (i.e., never or infrequent relapses vs. frequent relapses or steroid dependence; as defined by the 2023 IPNA guidelines [6]). Secondary outcomes among children with steroid-sensitive nephrotic syndrome were as follows: (1) disease relapse occurrence within 1 year of diagnosis and (2) initiation of a steroid-sparing immunosuppressive medication within 2 years of diagnosis. Among all children (steroid-sensitive and steroid-resistant), we evaluated steroid-resistant nephrotic syndrome classification after 4–6 weeks of initial steroid treatment at an appropriate dose. Secondary (i.e., late) steroid resistance was not included in this outcome definition.

Data preparation

Data exploration was performed to evaluate missing data and data distributions for clinically implausible values. We reported baseline characteristics of the overall population using descriptive statistics. Pairwise associations between candidate predictors were assessed using correlation matrices and pair plots. For collinearity, we examined generalized variance inflation factors (VIF) to evaluate redundancy among overlapping variables. To avoid information leakage, all preprocessing steps (imputation, scaling, encoding, and oversampling) were implemented within the training data only and applied to the held-out test data using parameters learned from the training set. The dataset was randomly split into training (70%) and testing (30%) sets and the same split was used for all models. On the split training and testing datasets separately, missing data (Table 1) was imputed using multiple

imputation by chained equations (with 10 iterations). One-hot encoding was used to convert categorical variables to numeric for machine learning models. Vitamin D level was dropped from all models due to high missingness (96%).

Since class imbalance was present for all primary and secondary outcomes (i.e., an unequal number of participants with and without the outcome; Table 1), synthetic minority oversampling technique (SMOTE) was performed on the split training sets only. SMOTE generates additional synthetic samples for the minority class using data from minority instances and their nearest neighbors to create balanced classes. For steroid-resistant nephrotic syndrome classification (the outcome with the largest class imbalance), we compared SMOTE oversampling with different class imbalance handling methods. These included random undersampling (random removal of majority class observations to create smaller, balanced training dataset), class weighting (assigning each observation an inverse frequency weight to penalize misclassification of minority class observations more heavily during model training), and stabilized inverse probability weighting (estimating the probability of each observation belonging to its observed class given its covariates, then re-weighting observations based on this probability with truncation of extreme weights at the 99th percentile).

Model development

We created supervised learning models for each primary and secondary outcome using the following: (1) logistic regression, (2) logistic regression with recursive feature elimination, (3) L2 (ridge) penalized logistic regression, (4) categorical naïve Bayes classifier with smoothing, (5) decision tree classifier, (6) random forest classifier, (7) XGBoost classifier, (8) deep learning neural network, and (9) support vector machine. Nine different algorithms, each with unique strengths and limitations, were tested to evaluate comparative model performance for each outcome (Supplemental Table 2). For logistic regression, neural network, and support vector machine models, predictor variables in the training and test datasets were rescaled and standardized to have a mean of 0 and variance of 1. For each model, hyperparameter tuning was performed using a grid search method with five-fold cross-validation.

Feature selection

We included all predictor variables in the machine learning models. For logistic regression model two, recursive feature elimination with three-fold cross-validation was used to select features, to evaluate if this improved model performance over standard logistic regression.

Table 1 Participant characteristics of 515 children enrolled in INSIGHT within 90 days of nephrotic syndrome diagnosis

Participant characteristic	Overall cohort (N = 515)	Missing data, n (%)
Demographic characteristics		
Male sex, n (%)	325 (63%)	0
Age at diagnosis in years, median (IQR)	3.9 (2.8–6.7)	0
Body mass index at diagnosis in kg/m ² , median (IQR)	17.4 (16.2–19.0)	19 (4%)
Rural residence, n (%)	22 (5%)	21 (4%)
Ethnicity, n (%)	119 (24%)	23 (5%)
European	172 (35%)	
South Asian	57 (12%)	
East/Southeast Asian	12 (2%)	
Middle Eastern	25 (5%)	
Caribbean	21 (4%)	
African	73 (15%)	
Mixed	11 (2%)	
Other ^a		
Immigrant to Canada, n (%)	70 (14%)	29 (6%)
Parental education, n (%)	29 (6%)	28 (5%)
Less than high school	186 (38%)	
High school	154 (32%)	
Undergraduate degree	118 (24%)	
Graduate degree		
Parental employment, n (%)	318 (74%)	84 (16%)
Total family income, n (%)	107 (30%)	159 (31%)
< \$50,000	117 (33%)	
\$50,000–99,999	132 (37%)	
≥ \$100,000		
Number of children living in the household, median (IQR)	2 (2–3)	91 (18%)
Birth characteristics		
Full-term delivery, n (%)	410 (86%)	38 (7%)
Birthweight in grams, median (IQR)	3260 (2870–3629)	149 (29%)
Family history		
Nephrotic syndrome, n (%)	14 (3%)	0
Kidney disease, n (%)	134 (26%)	0
Presenting symptoms		
Any symptoms, n (%)	503 (98%)	0
Facial edema, n (%)	417 (81%)	0
Peripheral edema, n (%)	327 (64%)	0
Abdominal distension, n (%)	236 (46%)	0
Weight gain, n (%)	73 (14%)	0
Low urine output, n (%)	28 (5%)	0
Viral illness in 6 months prior to diagnosis, n (%)	157 (40%)	122 (24%)
Infection treated with antibiotics in 6 months prior to diagnosis, n (%)	131 (33%)	121 (24%)
Hospitalization at diagnosis, n (%)	107 (22%)	33 (6%)
Intravenous albumin administration at diagnosis, n (%)	164 (36%)	56 (11%)
Atopic history		
Asthma, n (%)	64 (15%)	89 (17%)
Seasonal allergies, n (%)	26 (6%)	96 (19%)
Atopic dermatitis, n (%)	128 (30%)	92 (18%)
ISAAC score, median (IQR)	1 (0–1)	85 (17%)
Blood pressure		
Systolic blood pressure percentile, median (IQR)	94 (82–99)	30 (6%)
Diastolic blood pressure percentile, median (IQR)	95 (83–99)	37 (7%)
Laboratory measures		
Hemoglobin in g/L, mean (SD)	132 (15)	62 (12%)

Table 1 (continued)

Participant characteristic	Overall cohort (N = 515)	Missing data, n (%)
Serum sodium in mmol/L, mean (SD)	135 (4)	56 (11%)
Serum urea in mmol/L, mean (SD)	6.1 (4.4)	62 (12%)
Serum albumin in g/L, mean (SD)	19 (5)	53 (10%)
Serum cholesterol in mmol/L, mean (SD)	10.9 (3.1)	127 (25%)
Serum total 25-OH vitamin D in nmol/L, mean (SD)	12.9 (5.4)	494 (96%)
Urine albumin to creatinine ratio in mg/mmol, mean (SD)	790 (981)	271 (53%)
Urine protein to creatinine ratio in mg/mmol, mean (SD)	1433 (2126)	216 (42%)
Estimated glomerular filtration rate in mL/min/1.73 m ² , mean (SD)	117 (40)	113 (22%)
Hematuria present, n (%)	185 (36%)	40 (8%)
Proteinuria in g/L, n (%)	1 (< 1%)	30 (6%)
Negative	16 (3%)	
Trace	26 (5%)	
0.3	38 (7%)	
1	404 (79%)	
≥ 3		
Outcomes		
Frequently relapsing or steroid-dependent nephrotic syndrome by 1 year, n (%)	197/484 (41%)	0
Relapse occurrence by 1 year, n (%)	356/484 (74%)	0
Steroid-sparing medication initiation by 2 years, n (%)	217/484 (45%)	0
Initial steroid resistance, n (%)	31/515 (6%)	0

SD Standard deviation, IQR Interquartile range, ISAAC International Study of Asthma and Allergies in Childhood

^aLatin American, South American, or Indigenous ethnicity

Model performance

We evaluated performance using confusion matrices accuracy, precision, recall, and F1 score (as the macro average giving equal weight to each class). For all outcomes, models generated predicted probabilities, which were converted to class labels using a fixed threshold of 0.5 to ensure consistent, prespecified classification across models. We determined model fit and discrimination using ROC-AUC curves. We estimated 95% confidence intervals for AUC values through bootstrapping with 1000 resamples. We evaluated performance for imbalanced binary outcomes using precision–recall curves with predicted probabilities assessed across all classification thresholds. Model calibration (i.e., agreement between observed and predicted outcomes) was evaluated using Brier scores and calibration plots. For the primary outcome model with optimal performance, Shapley (SHAP) values were calculated to evaluate and visualize relative feature importance.

Results

Participant characteristics

Of 752 children diagnosed with nephrotic syndrome and enrolled in the INSIGHT study, 711 were classified as

steroid-sensitive and 41 were steroid-resistant. Of these, 515 (484 steroid-sensitive and 31 steroid-resistant) had their initial study visit within 90 days of nephrotic syndrome diagnosis and were included in this analysis (Table 1). The initial visit occurred at median (IQR) 7 (2–21) days after nephrotic syndrome diagnosis. Median (IQR) age at diagnosis was 3.9 (2.8–6.7) years, 325 (63%) of children were male, 119 (24%) were of European ethnicity, and 172 (35%) were of South Asian ethnicity. Symptoms were present at diagnosis in 503 (98%) children, most commonly facial and peripheral edema. Laboratory data from within 2 weeks of diagnosis was missing for vitamin D in 494 children (96%), urine albumin to creatinine ratio in 271 (53%), and urine protein to creatinine ratio in 216 (42%). Missingness for other laboratory variables ranged from 6 to 25% (Table 1). No clinically implausible values were identified for any predictor variables. All included predictor variables had generalized VIF values ≤ 3.5 , indicating a lack of substantial collinearity. Median (IQR) follow-up duration for the analysis cohort was 4.5 (2.6–7.9) years; 466 children (91%) had follow-up ≥ 1 year and 417 (81%) had follow-up ≥ 2 years.

Predicting frequent relapses or steroid dependence

Among 484 children with steroid-sensitive nephrotic syndrome, 197 (41%) were classified with frequently relapsing or steroid-dependent disease by 1 year after diagnosis. Nine

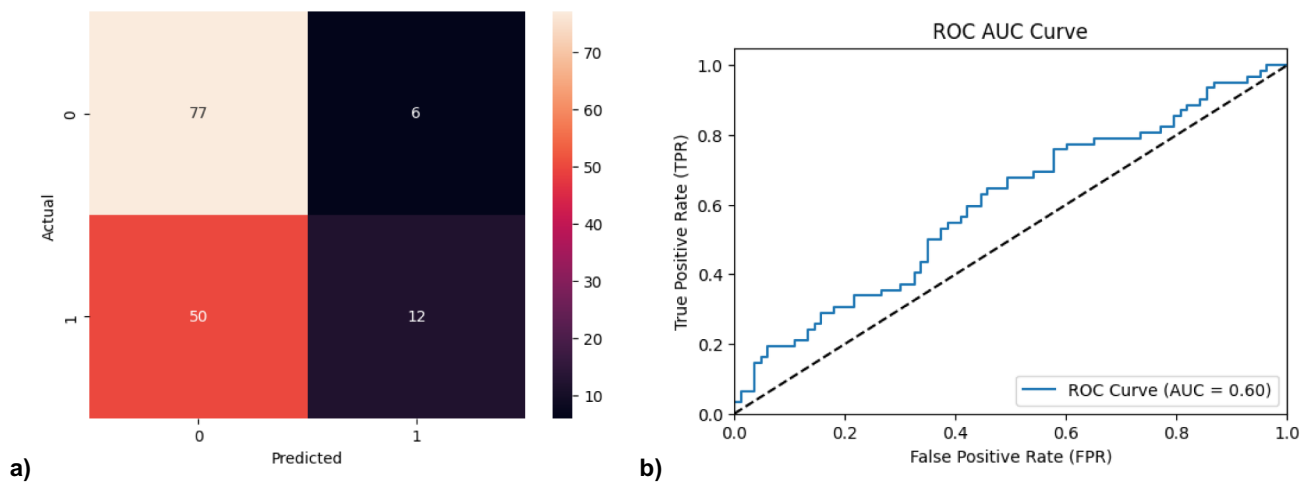


Fig. 1 Random forest model **a** classification matrix and **b** receiver operator curve for frequently relapsing or steroid-dependent nephrotic syndrome by 1 year after diagnosis. **a** shows the classification (i.e., confusion) matrix for the random forest classifier model for frequently relapsing or steroid-dependent nephrotic syndrome classification by 1 year after diagnosis. The purple box (predicted=1 and

actual=1) represent true positives. The cream box (predicted=0 and actual=0) represent true negatives. The orange box (predicted=0 and actual=1) represent false negatives. The black box (predicted=1 and actual=0) represent false positives. **b** shows the receiver operator curve (ROC) for this prediction model, which was found to have an AUC of 0.60 (low discriminative power)

machine learning models were developed to predict this outcome (Supplemental Table 3). Of these, the random forest classifier model was found to perform best for predicting frequently relapsing or steroid-dependent disease (Table 2). However, this model had poor discriminative ability (ROC-AUC=0.60 [95%CI 0.48–0.67], F1=0.52; Fig. 1), precision across different classification thresholds (Supplemental Fig. 1a), and calibration (Brier score=0.24; Supplemental Fig. 2a). Confidence intervals for AUCs were overlapping, demonstrating that no single model was superior. Feature importance varied by model. Recursive feature elimination for logistic regression identified eight variables for inclusion in the final model: birthweight, facial edema at presentation, a history of recent viral illness, ISAAC atopy score, diagnosis month, diastolic blood pressure percentile, cholesterol, and urine protein to creatinine ratio. For the random forest model, age at diagnosis, cholesterol level, and systolic blood pressure percentile were the most influential features, based on their SHAP values (Fig. 2).

Predicting steroid resistance, relapse, and steroid-sparing medication initiation

Among all 515 children with nephrotic syndrome, 31 (6%) were classified as steroid-resistant after their initial steroid treatment course. Of the nine machine learning models tested (Supplemental Table 4), the XGBoost classifier was the best performing (Table 2), although its discriminative ability was low (ROC-AUC=0.64 [95%CI 0.42–0.85], F1=0.49). Comparing different class imbalance handling methods, stabilized inverse probability weighting

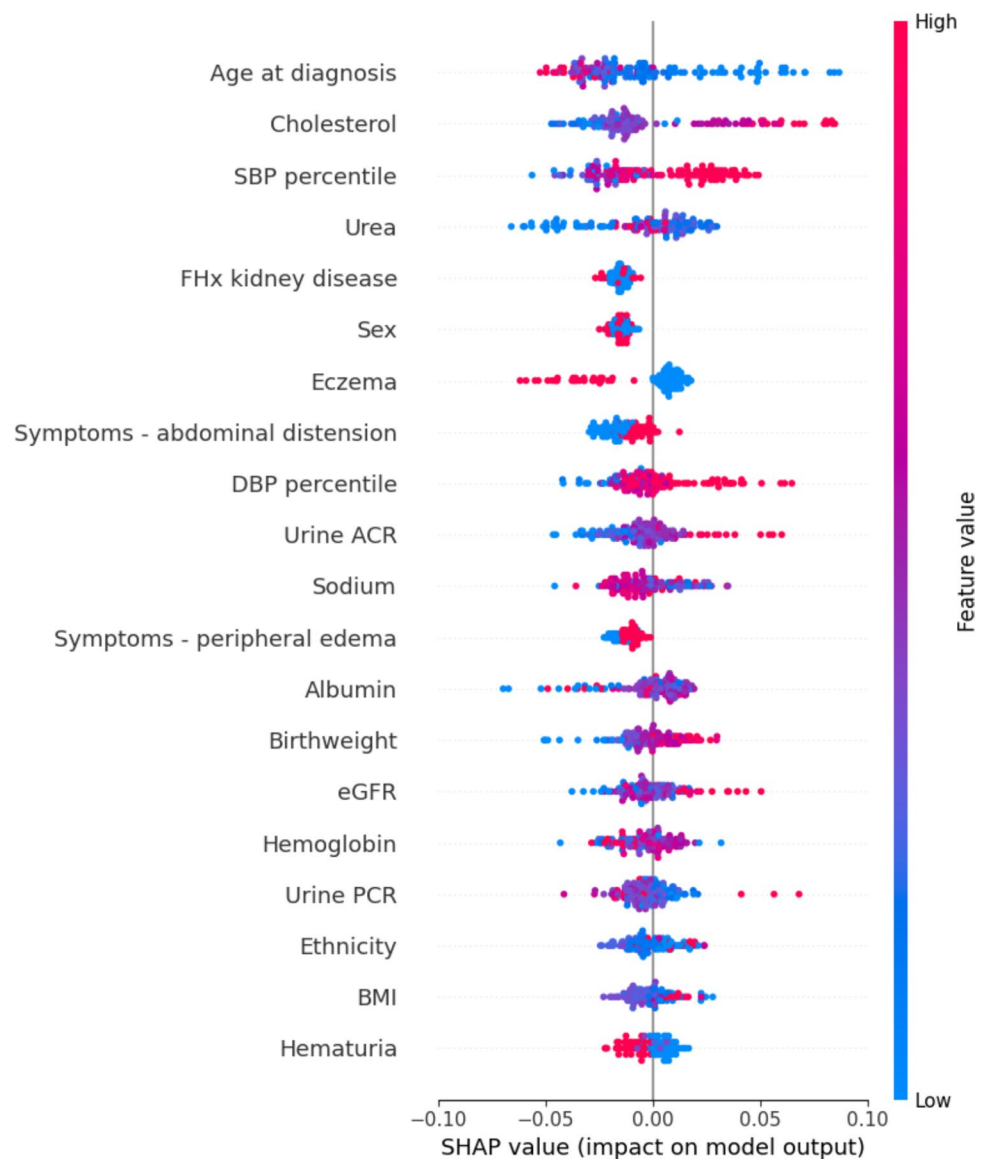
(ROC-AUC=0.73) and SMOTE oversampling (ROC-AUC=0.64) produced models with better discriminative ability than class weighting (ROC-AUC=0.58) or random undersampling (ROC-AUC=0.44; Supplemental Table 5). Among 484 children with steroid-sensitive nephrotic syndrome, 356 (74%) children experienced at least one nephrotic syndrome relapse by 1 year after diagnosis. By 2 years after diagnosis, 217 (45%) had initiated a steroid-sparing medication. For predicting relapse occurrence by 1 year after diagnosis (Supplemental Table 6), the logistic regression model with recursive feature elimination had the best performance (ROC-AUC=0.62 [95%CI 0.51–0.71], F1=0.54). For predicting steroid-sparing medication initiation by 2 years after diagnosis (Supplemental Table 7), the XGBoost classifier had the best performance (ROC-AUC=0.61 [95%CI 0.52–0.71], F1=0.57).

Precision across different classification thresholds (Supplemental Fig. 1) and calibration were poor for each of these models (Brier scores=0.06–0.32; Supplemental Fig. 2). Evidence of overfitting was found for all machine learning models, with substantially better model performance on training datasets than testing datasets (Supplemental Table 8).

Discussion

Nephrotic syndrome often follows a variable, unpredictable course, which can be frustrating for patients, caregivers, and healthcare providers. Currently, there are a lack of clinical features or biomarkers that can reliably predict

Fig. 2 Random forest model SHAP analysis for frequently relapsing or steroid-dependent nephrotic syndrome by 1 year after diagnosis. The figure shows Shapley (SHAP) values for the 20 most influential predictors on the output of the random forest classifier model for frequently relapsing or steroid-dependent nephrotic syndrome by 1 year after diagnosis. Predictors are ordered from most (top) to least (bottom) influential. Points further to the right are associated with a higher risk of the outcome and points further to the left are associated with a lower risk. Points shaded in red represent a higher value (e.g., older age at diagnosis) and points shaded in blue represent a lower value (e.g., younger age at diagnosis). For example, age at diagnosis was found to be the most influential predictor of frequently relapsing or steroid-dependent nephrotic syndrome. Younger age at diagnosis (represented by blue points) was associated with a higher risk of this outcome



key disease outcomes such as initial steroid response, subsequent relapses, or the need for steroid-sparing immunosuppressive medications. This study evaluated if a wide range of patient sociodemographic, clinical, and laboratory features that are routinely collected at the time of diagnosis could predict subsequent disease outcomes using machine learning techniques. The predictive ability of these models for frequently relapsing or steroid-dependent disease by 1 year after diagnosis was poor. This offers limited value to clinicians aiming to predict which children are more likely to experience frequent relapses and potentially benefit from maintenance immunosuppressive medications. Machine learning prediction models were also unable to adequately discriminate those at risk for initial steroid resistance, relapse occurrence by 1 year, and steroid-sparing medication initiation by 2 years. These findings suggest that

routinely collected sociodemographic, clinical, and laboratory features have limited ability to predict childhood nephrotic syndrome outcomes. Further studies are needed to identify and evaluate the predictive ability of novel biomarkers (e.g., genetic polymorphisms, autoantibodies, immune markers) that could be incorporated into future risk prediction models to improve their performance.

Previous studies report that younger age at diagnosis, male sex, and early disease activity are predictive of subsequent nephrotic syndrome relapses [11–16]. However, these associations are inconsistently observed. In a study of 129 Australian children, male sex and a longer time-to-remission during the initial nephrotic syndrome episode were associated with a higher risk of relapses [16]. Another study of 120 French children found that male sex, younger age at diagnosis, intravenous steroid treatment, and longer

Table 2 Best machine learning model performance for each outcome among children with nephrotic syndrome

Model	Accuracy	Precision	Recall	F1-score	Brier score	ROC-AUC
Frequently relapsing or steroid-dependent nephrotic syndrome by 1 year						
Random forest classifier ^a	0.61	0.64	0.56	0.52	0.24	0.60 (0.48–0.67)
Relapse occurrence by 1 year						
Logistic regression with recursive feature elimination ^b	0.56	0.56	0.58	0.54	0.26	0.62 (0.51–0.71)
Steroid-sparing medication initiation by 2 years						
XGBoost classifier ^c	0.57	0.57	0.57	0.57	0.32	0.61 (0.52–0.71)
Steroid-resistant nephrotic syndrome after initial steroid treatment						
XGBoost classifier ^d	0.94	0.47	0.50	0.49	0.06	0.64 (0.42–0.85)

For each outcome, nine machine learning models were developed: (1) logistic regression, (2) logistic regression with recursive feature elimination, (3) L2 (ridge) penalized logistic regression, (4) categorical naïve Bayes classifier with smoothing, (5) decision tree classifier, (6) random forest classifier, (7) XGBoost classifier, (8) deep learning neural network, and (9) support vector machine. Hyperparameter tuning was performed for each model by grid search to maximize ROC-AUC. For each outcome, the best performing model was selected based on the highest ROC-AUC value

^aOptimal hyperparameters were as follows: 50 decision trees, maximum tree depth of 20, minimum five samples to split an internal node, and minimum four samples at each leaf node

^bTwelve features were selected by recursive feature elimination for the final model (age at diagnosis, immigration status, facial edema at presentation, weight gain at presentation, oliguria at presentation, history of recent viral illness, diagnosis month, intravenous albumin use at diagnosis, diastolic blood pressure percentile, body mass index, urine albumin to creatinine ratio, and urine protein to creatinine ratio)

^cOptimal hyperparameters were as follows: a learning rate of 0.1, a maximum tree depth of seven, 200 boosting rounds, and 100% of training data used for each tree

^dOptimal hyperparameters were as follows: a learning rate of 0.01, a maximum tree depth of five, 200 boosting rounds, and 80% of training data used for each tree

time-to-remission were associated with increased subsequent steroid-sparing medication use [23]. Among 631 Canadian children diagnosed with nephrotic syndrome between 1993 and 2016 and enrolled in INSIGHT, prediction models developed by logistic regression including different combinations of sociodemographic and clinical features were poorly discriminative for frequent relapses, steroid-sparing medication use, and long-term remission (*c*-statistics ranging from 0.50 to 0.67) [4]. We attempted to build upon this previous work by including additional sociodemographic, clinical, and laboratory features and developing prediction models using multiple machine learning methods. Despite these advantages over previous prediction models, we still observed that information from the time of nephrotic syndrome diagnosis was weakly predictive of subsequent outcomes. Consistent with prior studies, we found that younger age at diagnosis was associated with frequent relapses or steroid dependence.

Few studies have evaluated the predictive ability of machine learning models among children with nephrotic syndrome. For predicting steroid resistance, machine learning models have been recently developed with moderate-to-strong predictive ability. A recent study of 243 Iranian children reported that a random forest classifier model had high predictive ability for steroid resistance (ROC-AUC 0.80), steroid dependence (ROC-AUC 0.87),

and frequent relapses (ROC-AUC 0.99) [17]. However, this cohort included infants < 1 year (15% of the cohort), those with congenital anomalies (18%), and those with secondary forms of nephrotic syndrome (26%). As a result, 59% of cases were steroid-resistant, which is considerably higher than the 5–10% prevalence of steroid resistance reported in most childhood nephrotic syndrome cohorts and may not be generalizable to other populations [4, 24, 25]. Additionally, this study estimated model performance using the same cross-validation sets that were used for data pre-processing and hyperparameter tuning, which would result in information leakage and overly optimistic performance estimates. Another single center study of 91 Chinese children found that support vector machine models including 27 clinical and laboratory features, several of which are not routinely measured in clinical practice, had a strong predictive ability (ROC-AUC 0.98) for steroid resistance [18]. Interestingly, a reduced model including only eight laboratory features (including vinculin autoantibodies) achieved a similar predictive ability (ROC-AUC 0.95). The performance of these models was likely substantially improved by the inclusion of specific laboratory and autoantibody features. For example, vinculin autoantibodies were identified from 17 candidate autoantibodies to be strongly associated with steroid resistance. A multicenter study

of 329 Japanese children and young adults with nephrotic syndrome found that machine learning models including proteinuria change within the first 2 weeks of initial steroid therapy was strongly predictive (ROC-AUC > 0.9) of steroid resistance and moderately predictive (ROC-AUC 0.72–0.76) of steroid-sparing medication initiation by 2 years [19]. Another study of 124 Polish children with nephrotic syndrome found that a neural network including multiple genetic features was moderately predictive of steroid resistance (ROC-AUC 0.78) and steroid dependence (ROC-AUC 0.72) [20]. Compared with these studies, we included a substantially larger cohort, incorporated additional predictor features, and trained models using methods to prevent information leakage. We purposively selected predictor features that would be routinely collected during the initial nephrotic syndrome episode, to evaluate their predictive ability for subsequent outcomes. This would facilitate external validation of these models using data from existing international cohorts and implementation in routine clinical care. Using this approach, our best-performing model for steroid resistance, the XGBoost classifier, achieved an ROC-AUC of only 0.64, lower than prior studies. This discrepancy suggests that incorporating specific immune biomarkers, autoantibodies, genetics, and serial laboratory measures may be needed to increase predictive model performance.

Machine learning techniques have been applied to other pediatric and adult glomerular diseases and outcomes with encouraging results. For example, among hospitalized children with nephrotic syndrome, a light gradient boosting machine learning model was found to be highly predictive of severe infections (ROC-AUC 0.91) [26]. An XGBoost model to predict 5-year progression to kidney failure among 1146 children with IgA nephropathy achieved a ROC-AUC of 0.85 [27]. Machine learning methods have also been used to predict adverse kidney events in adults with idiopathic membranous nephropathy (ROC-AUC 0.89) [28]. Among 248 adults with nephrotic syndrome, machine learning algorithms including clinical and laboratory features were found to be strongly predictive (ROC-AUC 0.92) of minimal change disease histopathological diagnosis [29]. These studies demonstrate several potential uses for machine learning techniques to improve prediction of glomerular disease outcomes and limit unnecessary procedures such as kidney biopsy. Additionally, the application of deep learning methods to histological image analysis can improve disease classification and prognostication [30, 31]. Predicting nephrotic syndrome relapses or treatment response may be inherently more complex, since many unknown genetic and environmental factors likely influence these outcomes.

Our findings have several important implications for clinical practice and future research. First, clinical

decision-making should remain guided by observed patient response and course, until accurate prediction models are developed. These results also highlight the urgent need for the discovery of novel biomarkers that predict nephrotic syndrome course. The weak performance of models in this study suggests that routinely collected clinical and laboratory data, such as age at diagnosis, are insufficient to predict nephrotic syndrome outcomes and should not be used when counselling families. Strengths of this study include the large, ethnically diverse, multi-center cohort from which detailed patient demographic, clinical, and laboratory data was collected. Multiple machine learning algorithms were developed for each outcome, trained using many predictor variables, and algorithms underwent hyperparameter tuning to optimize performance. However, this analysis also has several limitations. We observed a high degree of missingness for several variables included in our models, particularly certain laboratory measures (e.g., urine protein to creatinine [42%] and albumin to creatinine ratio [53%]). This reduced effective sample size for laboratory predictors and may degrade model training. This was addressed by multiple imputation, which could introduce bias since missingness may not be at random. Other non-parametric imputation techniques, such as k-nearest neighbors or tree-based imputation, and feature extraction methods could have improved model performance. However, the low discriminative ability of all models suggests that standard laboratory measures at diagnosis may not be strong predictors of subsequent nephrotic syndrome outcomes. Predictor variables that would only be available after diagnosis (e.g., time-to-remission) were excluded, since our aim was to develop prediction models to be used at the time of diagnosis. Inclusion of additional post-diagnosis features may improve model performance and warrants further study. Given the prolonged 27-year enrollment period, associations between predictor variables and outcomes may have changed over time, particularly for steroid-sparing medication initiation. Overfitting also appears to be an issue, since model performance was considerably better on training datasets than testing datasets. Various strategies were used to limit overfitting, including model penalization, oversampling, and neural network early stopping. Despite these measures, some overfitting clearly remains since performance drops were observed moving from training to testing sets. Other strategies, such as restricting feature selection and nested cross-validation, could have been used to further reduce overfitting. For steroid resistance, low outcome prevalence (6%) limits the ability of machine learning models to detect complex predictive patterns, which cannot be fully addressed through class imbalance handling methods. Finally, our prediction models were internally validated using a split testing set from the same

cohort, which may inflate model performance. Since model performance was poor, external validation on other nephrotic syndrome cohorts was not performed.

Conclusions

Routinely collected sociodemographic, clinical, and laboratory data from the time of childhood nephrotic syndrome diagnosis were poorly predictive of steroid resistance, subsequent relapse patterns, and initiation of steroid-sparing medications using machine learning methods. These results are important for patient counselling, emphasizing that we currently lack the ability to accurately predict disease course among children diagnosed with nephrotic syndrome based on available markers. This is also a strong justification for future research to identify and validate the predictive ability of novel factors, such as genetic, autoantibody, or immune response markers.

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Author contribution Cal Robinson designed the study, performed all data cleaning and statistical analyses, produced the tables and figures, and drafted the manuscript. Tanay Joshi critically appraised and revised the study protocol, performed literature review, and drafted the manuscript. Nowrin Aman, Tonny Banh, Mackenzie Garner, and Jovanka Vasilevska-Ristovska contributed to data collection and critically appraised and revised the study protocol and initiation documents. Konrad Samsel, Zahra Shakeri, Josefina Brooke, Vaneet Dhillon, Christoph Licht, Ashlene McKay, Rachel Pearl, Seetha Radhakrishnan, Keisha Rasool, Nithiakishna Selvathesan, and Chia Wei Teoh critically appraised and revised the study protocol and initiation documents and revised the draft manuscript. Rulan Parekh designed the study, provided supervision, and revised the draft manuscript. All authors approved the final manuscript as submitted and agree to be accountable for all aspects of the work.

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Data availability Deidentified individual participant data will not be made available. All Python code used to perform this analysis is available online in the following GitHub repository: https://github.com/cal-robinson/INSIGHT_ML_Prediction.

Declarations

Disclosures All authors declare no real or perceived conflicts of interest that could affect the study design, collection, analysis and interpretation of data, writing of the report, or the decision to submit for publication.

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